Deqing Fu

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Google Scholar

Research Interests

I study the **theoretical foundations of large language models** and ways to interpret, align, and extend them beyond text.

- **Algorithmic perspectives on LLMs**: whether Transformers learn algorithms from data [C4, C14], how pretrained LLMs do arithmetic [C3, C10, C18], and what sets Transformers apart from other architectures [C7].
- **Interpretability and alignment**: decision making under uncertainty [C6, C15], steering vectors [C12, C13], faithfulness via token-level rewards [C17], and mechanistic interpretability with SAEs and transcoders [C16, T1].
- **Foundation models beyond text**: multimodal reward models [C8, C5], modality sensitivity [C2], data for visual reasoning [C11], and tabular foundation models [T2, P4].

Education

University of Southern California 2022 – Present
Doctor of Philosophy (Ph.D.) in Computer Science
Advisors: Vatsal Sharan & Robin Jia

University of Chicago 2020 – 2022
Master of Science (S.M.) in Statistics

University of Chicago 2016 – 2020
Bachelor of Science (S.B.) in Mathematics with Honors
Bachelor of Science (S.B.) in Computer Science with Honors
Bachelor of Arts (A.B.) in Statistics

Experience

Google Research May 2026 – Present
Student Researcher • Hosts: Weihao Kong, Abhimanyu Das

Simons Institute at UC Berkeley Fall 2024
Visiting Graduate Student • Program: Modern Paradigms in Generalization

Meta Llama Team May – Dec., 2024
AI Research Scientist Intern • Host: Lawrence Chen; Peers: Tong Xiao, Rui Wang, and Pengchuan Zhang

Google May – Aug., 2023
Software Engineering Intern • Team: Bard (now Gemini) Multilingual

Publications

featured * equal contribution † mentored student

[C19] **Value-Aware Stochastic KV Cache Eviction for Reasoning Models**
Ting-Yun Chang*, Harvey Yiyun Fu*, Deqing Fu*, Chenghao Yang, Jesse Thomason, Robin Jia
Neural Information Processing Systems (NeurIPS), 2026

[C18] **Convergent Evolution: How Different Language Models Learn Similar Number Representations**
Deqing Fu, Tianyi Zhou, Mikhail Belkin, Vatsal Sharan, Robin Jia
Conference on Language Modeling (COLM), 2026

[C17] **Token-Level Off-Policy Learning for Faithful Generation Under Distribution Shift**
Cynthia Huang*†, Gustavo Lucas Carvalho*†, Deqing Fu, Robin Jia
Conference on Language Modeling (COLM), 2026

[C16] **Resa: Transparent Reasoning Models via SAEs**
Shangshang Wang, Julian Asilis, Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, Deqing Fu, Willie Neiswanger
Conference on Language Modeling (COLM), 2026

[C15] **Are LLM Decisions Faithful to Verbal Confidence?**
Jiawei Wang, Yanfei Zhou, Siddartha Devic, Deqing Fu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026

[C14] **Transformers Provably Learn Algorithmic Solutions for Graph Connectivity, But Only with the Right Data**
Qilin Ye*†, Deqing Fu*, Robin Jia, Vatsal Sharan
International Conference on Machine Learning (ICML), 2026

[C13] **EPSVec: Efficient and Private Synthetic Data Generation via Dataset Vectors**
Amin Banayeeanzade, Qingchuan Yang, Deqing Fu, Spencer Hong, Erin Babinsky, Alfy Samuel, Anoop Kumar, Robin Jia, Sai Praneeth Karimireddy
International Conference on Machine Learning (ICML), 2026

[C12] **Textual Steering Vectors Can Improve Visual Understanding in Multimodal Large Language Models**
Woody Haosheng Gan*†, Deqing Fu*, Julian Asilis*, Ollie Liu*, Vatsal Sharan, Robin Jia, Willie Neiswanger
Association for Computational Linguistics (ACL), 2026

[C11] **Zebra-CoT: A Dataset for Interleaved Vision Language Reasoning**
Ang Li*, Charles L. Wang*, Deqing Fu*, Kaiyu Yue*, Zikui Cai*, Wang Bill Zhu*, Ollie Liu*, Peng Guo*, Willie Neiswanger, Furong Huang, Tom Goldstein, Micah Goldblum
International Conference on Learning Representations (ICLR), 2026

[C10] **FoNE: Precise Single-Token Number Embeddings via Fourier Features**
Tianyi Zhou, Deqing Fu, Mahdi Soltanolkotabi, Robin Jia, Vatsal Sharan
International Conference on Learning Representations (ICLR), 2026

[C9] **VisualLens: Personalization through Task-Agnostic Visual History**
Wang Bill Zhu, Deqing Fu, Kai Sun, Yi Lu, Zhaojiang Lin, Seungwhan Moon, Kanika Narang, Mustafa Canim, Yue Liu, Anuj Kumar, Xin Luna Dong
Neural Information Processing Systems (NeurIPS), 2025

[C8] **TLDR: Token-Level Detective Reward Model for Large Vision Language Models**
Deqing Fu, Tong Xiao, Rui Wang, Wang Zhu, Pengchuan Zhang, Guan Pang, Robin Jia, Lawrence Chen
International Conference on Learning Representations (ICLR), 2025

[C7] **Transformers Learn Low Sensitivity Functions: Investigations and Implications**
Bhavya Vasudeva*, Deqing Fu*, Tianyi Zhou, Elliot Kau†, You-Qi Huang†, Vatsal Sharan
International Conference on Learning Representations (ICLR), 2025

[C6] **DeLLMa: Decision Making Under Uncertainty with Large Language Models**
Ollie Liu*, Deqing Fu*, Dani Yogatama, Willie Neiswanger
International Conference on Learning Representations (ICLR), 2025 — ICLR 2025 Spotlight (Top 5.1%)

[C5] **DreamSync: Aligning Text-to-Image Generation with Image Understanding Feedback**
Jiao Sun*, Deqing Fu*, Yushi Hu*, Su Wang, Royi Rassin, Da-Cheng Juan, Dana Alon, Charles Herrmann, Sjoerd van Steenkiste, Ranjay Krishna, Cyrus Rashtchian
Nations of the Americas Chapter of the ACL (NAACL), 2025

[C4] **Transformers Learn to Achieve Second-Order Convergence Rates for In-Context Linear Regression**
Deqing Fu, Tian-Qi Chen, Robin Jia, Vatsal Sharan
Neural Information Processing Systems (NeurIPS), 2024 — SoCalNLP Symposium 2023 Best Paper Award

[C3] **Pre-trained Large Language Models Use Fourier Features to Compute Addition**
Tianyi Zhou, Deqing Fu, Vatsal Sharan, Robin Jia
Neural Information Processing Systems (NeurIPS), 2024

[C2] **IsoBench: Benchmarking Multimodal Foundation Models on Isomorphic Representations**
Deqing Fu*, Ruohao Guo*, Ghazal Khalighinejad*, Ollie Liu*, Bhuwan Dhingra, Dani Yogatama, Robin Jia, Willie Neiswanger
Conference on Language Modeling (COLM), 2024

[C1] **SCENE: Self-Labeled Counterfactuals for Extrapolating to Negative Examples**
Deqing Fu, Ameya Godbole, Robin Jia
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023

Pre-prints, Tech Reports, and Blog Posts

featured * equal contribution † mentored student

[P5] **TimeRSI: Towards Self-Improved Agentic Deployment for Time-Series Foundation Models**
Haoxin Liu, Deqing Fu, Lucheng Fu, Yiqiao Jin, Rushi Qiang, Harshavardhan Kamarthi, B. Aditya Prakash

[P4] **TabFM-Auto: Self-Evolving Pipelines for Tabular Foundation Models**
Deqing Fu, Huangyuan Su, Rajat Sen, Taman Narayan, Sujay Sanghavi, Abhimanyu Das, Weihao Kong

[P3] **PreciseCoder: Faithful and Generalizable Debugging Beyond Unit-Test Rewards**
Yunkai Zhan, Miaosen Chai, Shangshang Wang, Jike Zhong, Yuqing Yang, Deqing Fu, Serina Chang, Wang Bill Zhu

[P2] **Learning to GUIDE the Group: Self-Learning LLM Facilitation Across Multi-Party Tasks**
Ziyi Liu, Tzu-Sheng Kuo, Jiale Liu, Deqing Fu, Nithum Thain, Crystal Qian

[P1] **Unsupervised Tool Evolution for Coding Agents in Data-Lake Analytics**
Yuqing Yang, Yurong Liu, Deqing Fu, Manav Garg, Sravan Babu Bodapati, Fatma Ozcan, Alon Halevy, Jiani Zhang

[T2] **TabFM: A Zero-shot Foundation Model for Tabular Data** *Technical Report*
Weihao Kong, Erez Louidor Ilan, Shuxin Nie, Taman Narayan, Rajat Sen, Yichen Zhou, Deqing Fu, Samet Oymak, Abhimanyu Das

[T1] **Qwen3 Cross-layer Transcoders** *Technical Report*
Jakob Hansen, Deqing Fu, Shangshang Wang, John Carlsson

[B1] **Seeing Is Not Reasoning: How VLMs and Their Benchmarks Lean on Text** *Blog Post*
Harvey Yiyun Fu*, Chenghao Yang*, Ting-Yun Chang, Zhaowei Wang, Zhaofeng Wu, Jiawei Zhou, Deqing Fu

Awards

ACL Outstanding Reviewer Award — ACL, 2026
NeurIPS Top Reviewer — NeurIPS, 2025
Viterbi Best Research Assistant Award — USC, 2025
One per department in the Viterbi School of Engineering
USC Capital One Center (CREDIF) PhD Fellow — USC, 2025
SoCalNLP Symposium 2023 Best Paper Award — SoCalNLP, 2023
Provost's Fellowship — USC, 2022
Susanne H. Rudolph Scholarship — UChicago, 2020–2021
Liew Family College Research Fellows Fund — UChicago, 2020
Dean's List — UChicago, 2016–2020

Invited Talks

Emergent Algorithms in Foundation Models: Theory, Discovery, and Design
Columbia NLP Seminar — Nov 5, 2026
Cornell Tech ML Seminar — Nov 3, 2026
Northeastern University Theory Seminar — Oct 30, 2026
Boston University — Oct 29, 2026
Harvard ML Foundations Seminar — Oct 28, 2026
ISI Natural Language Seminar — Sep 22, 2026
USC NLP Lunch — Sep 10, 2026
Google Research Seminar — Aug 6, 2026

Algorithmic Perspectives on Understanding Transformers
FlaNN Talk Seminar — Aug 25, 2025
UChicago/TTIC NLP Seminar — May 08, 2025
Duke NLP Seminar — Feb 12, 2025

Closing the Modality Gap: Benchmarking and Improving Visual Understanding in Multimodal LLMs
Stanford NLP Seminar — May 22, 2025

Services

Reviewer for conferences in

- **Machine Learning**: ICLR (2025–2027), NeurIPS (2024–2026), ICML (2025)
- **Natural Language Processing**: COLM (2024–2026), EMNLP (2023), ACL Rolling Review (2024–2026)
- **Computer Vision**: CVPR (2023–2025), ECCV 2022, ICCV 2025

Organizer of USC ML/NLP Joint Reading Group — Jan 2025 – May 2025

Organizer of USC AI Safety Club — Jan 2024 – Jan 2025

Students Supervised

Undergraduate Students

Zitong Cynthia Huang (USC UG) — Sep 2025 – May 2026
Now: Ph.D. Student in Electronic and Computer Engineering at University of Southern California

Qilin Ye (USC UG) — Sep 2023 – Feb 2026
Now: Ph.D. Student in Computer Science at Stanford University

Woody Haosheng Gan (USC UG) — Sep 2023 – May 2026
Now: Ph.D. Student in Computer Science at Georgia Institute of Technology

You-Qi Huang (USC UG) — Sep 2023 – Mar 2024

Master's Students

Sophia Mirrashidi (USC MS) — Jan 2026 – Present

Gustavo Lucas Carvalho (USC MS) — Jan 2025 – May 2026

Elliot Kau (USC MS) — Sep 2023 – Mar 2024
Now: Software Engineer at Google

Tian-Qi Chen (USC MS) — Mar 2023 – Sep 2023

Teaching Experience

Teaching Assistant, CSCI 467: Introduction to Machine Learning — Spring 2025
University of Southern California, Los Angeles, CA

Teaching Assistant, CSCI 566: Deep Learning and its Applications — Spring 2023
University of Southern California, Los Angeles, CA

Teaching Assistant, BUSN 41301: Statistical Insight — Fall 2021
Booth School of Business, University of Chicago, Chicago, IL

Course Assistant, CMSC 31230: Fundamentals of Deep Learning — Fall 2021
University of Chicago, Chicago, IL

Course Assistant, MATH 20300: Analysis in $\mathbb{R}^n$ – I — Spring 2018
University of Chicago, Chicago, IL

Skills

Languages: C/C++, Python, LaTeX, MATLAB, R, Java, CUDA
Frameworks: PyTorch, TensorFlow, Keras, JAX, HuggingFace